

Swiggy Business Analytics Case Study

Scenario

You have recently joined Swiggy as a Junior Data Analyst. Your manager has shared the customer, restaurant, and order databases. Your task is to analyze the data and answer the following business questions using SQL.

```
CREATE TABLE Customers (  
  CustomerID INT PRIMARY KEY,  
  CustomerName VARCHAR(50),  
  City VARCHAR(30),  
  Membership VARCHAR(20)  
);
```

```
INSERT INTO Customers VALUES  
(101, 'Amit Sharma', 'Delhi', 'Gold'),  
(102, 'Priya Verma', 'Mumbai', 'Silver'),  
(103, 'Rahul Singh', 'Delhi', 'Gold'),  
(104, 'Sneha Gupta', 'Pune', 'Regular'),  
(105, 'Karan Mehta', 'Bangalore', 'Silver'),  
(106, 'Anjali Jain', 'Delhi', 'Gold'),  
(107, 'Rohit Kumar', 'Hyderabad', 'Regular'),  
(108, 'Neha Kapoor', 'Mumbai', 'Gold'),  
(109, 'Pooja Arora', 'Pune', 'Silver'),  
(110, 'Vikas Malhotra', 'Chennai', 'Regular'),  
(111, 'Riya Sharma', 'Jaipur', 'Gold'),  
(112, 'Aditya Singh', 'Lucknow', 'Regular');
```

```
CREATE TABLE Restaurants (  
  RestaurantID INT PRIMARY KEY,  
  RestaurantName VARCHAR(50),  
  Cuisine VARCHAR(30),  
  City VARCHAR(30),  
  Rating DECIMAL(2,1)  
);
```

```
INSERT INTO Restaurants VALUES  
(201, 'Burger King', 'Fast Food', 'Delhi', 4.4),  
(202, 'Dominos', 'Pizza', 'Mumbai', 4.5),  
(203, 'Biryani Blues', 'Biryani', 'Delhi', 4.7),  
(204, 'Pizza Hut', 'Pizza', 'Pune', 4.3),  
(205, 'Subway', 'Healthy', 'Bangalore', 4.2),  
(206, 'KFC', 'Fast Food', 'Hyderabad', 4.1),  
(207, 'Haldiram's', 'North Indian', 'Delhi', 4.6),  
(208, 'Wow Momo', 'Chinese', 'Mumbai', 4.2),  
(209, 'Barbeque Nation', 'BBQ', 'Pune', 4.8),  
(210, 'Faasos', 'Rolls', 'Chennai', 4.0),  
(211, 'McDonalds', 'Fast Food', 'Delhi', 4.3),  
(212, 'Behrouz Biryani', 'Biryani', 'Bangalore', 4.7);
```

```
DROP Table orders;
```

```

CREATE TABLE Orders (
OrderID INT PRIMARY KEY,
CustomerID INT,
RestaurantID INT,
OrderAmount INT,
DeliveryPartner VARCHAR(30),
OrderDate DATE,
OrderStatus VARCHAR(20),
FOREIGN KEY(CustomerID) REFERENCES
Customers(CustomerID),
FOREIGN KEY(RestaurantID) REFERENCES
Restaurants(RestaurantID)
);

```

```

INSERT INTO Orders VALUES
(1001, 101, 201, 450, 'Rahul', '2026-08-01', 'Delivered'),
(1002, 102, 202, 780, 'Aman', '2026-08-01', 'Delivered'),
(1003, 103, 203, 620, 'Rohit', '2026-08-02', 'Delivered'),
(1004, 104, 204, 540, 'Rahul', '2026-08-02', 'Cancelled'),
(1005, 105, 205, 390, 'Aman', '2026-08-03', 'Delivered'),
(1006, 106, 207, 860, 'Neha', '2026-08-03', 'Delivered'),
(1007, 107, 206, 710, 'Rohit', '2026-08-04', 'Delivered'),
(1008, 108, 208, 580, 'Rahul', '2026-08-04', 'Delivered'),
(1009, 109, 209, 950, 'Aman', '2026-08-05', 'Delivered'),
(1010, 110, 210, 430, 'Neha', '2026-08-05', 'Delivered'),
(1011, 101, 203, 720, 'Rahul', '2026-08-06', 'Delivered'),
(1012, 102, 202, 810, 'Aman', '2026-08-06', 'Delivered'),
(1013, 103, 207, 650, 'Neha', '2026-08-07', 'Delivered'),
(1014, 104, 204, 490, 'Rohit', '2026-08-07', 'Delivered'),
(1015, 105, 212, 880, 'Rahul', '2026-08-08', 'Delivered'),
(1016, 106, 211, 520, 'Aman', '2026-08-08', 'Cancelled'),
(1017, 107, 206, 640, 'Neha', '2026-08-09', 'Delivered'),
(1018, 108, 208, 760, 'Rahul', '2026-08-09', 'Delivered'),
(1019, 109, 209, 990, 'Rohit', '2026-08-10', 'Delivered'),
(1020, 110, 210, 450, 'Aman', '2026-08-10', 'Delivered');

```

-- 1. How many total orders have been placed on the platform?

```

SELECT COUNT(*) as Total_orders
FROM orders;

```

-- 2.What is the total revenue generated by Swiggy?

```

SELECT SUM(OrderAmount) as Total_Revenue
FROM Orders;

```

-- 3. What is the average order value across all orders?

```

SELECT AVG(OrderAmount) as Avg_Order_Value
FROM Orders;

```

-- 4. Which are the Top 5 highest-value orders placed by customers?

```

SELECT (OrderAmount) From Orders
Order by OrderAmount DESC
LIMIT 5;

```

-- 5. Which city generated the highest revenue for Swiggy?

```
SELECT c.City,  
       SUM(o.OrderAmount) As Revenue  
FROM Orders o  
Join Customers c  
ON o.customerID = c.CustomerID  
Group By c.City  
Order By revenue DESC  
LIMIT 1;
```

-- 6. Which restaurant generated the highest total revenue?

```
SELECT r.RestaurantName,  
       SUM(o.OrderAmount) As Total_Revenue  
FROM Orders o  
Join Restaurants r  
ON r.restaurantID = o.restaurantID  
group by r.restaurantName  
order by Total_revenue DESC  
Limit 1;
```

-- 7. Which cuisine is the most popular based on the number of
-- orders received?

```
SELECT  
       r.Cuisine,  
       COUNT(*) AS Total_Orders  
FROM Orders o  
JOIN Restaurants r  
ON o.RestaurantID = r.RestaurantID  
GROUP BY r.Cuisine  
ORDER BY Total_Orders DESC  
LIMIT 1;
```

-- 8. Which delivery partner completed the highest number of
-- successful deliveries?

```
Select DeliveryPartner,  
COUNT(*) as successful_deliveries  
FROM Orders  
where OrderStatus = 'Delivered'  
group by DeliveryPartner  
order by successful_deliveries DESC  
limit 1;
```

-- 9. Which customers have spent more than the average order
-- value?

```
Select c.CustomerName,  
o.OrderAmount  
From orders o  
Join customers c  
on c.customerID = o.customerID
```

```
where o.OrderAmount >
(Select Avg(OrderAmount)
From Orders);
```

```
-- 10. Find the second highest order amount placed on the platform.
SELECT DISTINCT OrderAmount
FROM Orders
ORDER BY OrderAmount DESC
LIMIT 1 OFFSET 1;
```

```
-- 11. Which restaurant has the highest average customer rating?
Select restaurantName,
rating
From Restaurants
where Rating =
( Select Max(rating)
from restaurants);
```

```
-- 12. Which cities generated revenue greater than ₹2,000?
Select c.city,
SUM(OrderAmount) as Total_amount
From customers c
join orders o
on c.CustomerID = o.CustomerID
group by city
having SUM(o.OrderAmount) > 2000;
```

```
-- 13. Which customers have never placed an order?
Select c.customerID,
c.customerName
From customers c
left join orders o
on c.CustomerID = o.CustomerID
where o.orderID is null;
```

```
-- 14. Which restaurants have never received any orders?
select r.restaurantID,
       r.restaurantName
from restaurants r
right join orders o
on r.restaurantID = o.restaurantID
where orderID is null;
```

```
-- 15. Display each customer's order details, including Customer
-- Name, Restaurant Name, Cuisine, Order Amount, and Order
-- Status.
SELECT
```

```

c.CustomerName,
r.RestaurantName,
r.Cuisine,
o.OrderAmount,
o.OrderStatus
FROM Orders o
INNER JOIN Customers c
ON o.CustomerID = c.CustomerID
INNER JOIN Restaurants r
ON o.RestaurantID = r.RestaurantID;

```

-- 16. Categorize every order as Budget, Medium, or Premium based
-- on the order amount.

```

SELECT
    OrderID,
    OrderAmount,
    CASE
        WHEN OrderAmount <= 500 THEN 'Budget'
        WHEN OrderAmount BETWEEN 501 AND 800 THEN 'Medium'
        ELSE 'Premium'
    END AS Category
FROM Orders;

```

-- 17. Rank customers based on their total spending on Swiggy.

```

SELECT c.customer,
    SUM(o.OrderAmount) as Total_spending
    Rank() Over(
        Order by Sum(o.OrderAmount) DESC
    ) AS Customer_Rank
From customer c
Join orders o
on c.customerID = o.customerID

```

-- 18. Assign a unique serial number to every order based on the order
-- amount from highest to lowest.

```

SELECT
    OrderID,
    OrderAmount,
    ROW_NUMBER() OVER(
        ORDER BY OrderAmount DESC
    ) AS Serial_Number
FROM Orders;

```

-- 19. Rank restaurants based on the total revenue they generated. If
-- two restaurants have the same revenue, they should receive the
-- same rank without skipping the next rank.

```

SELECT
    r.RestaurantName,

```

```

SUM(o.OrderAmount) AS Revenue,
DENSE_RANK() OVER(
    ORDER BY SUM(o.OrderAmount) DESC
) AS Restaurant_Rank
FROM Restaurants r
JOIN Orders o
ON r.RestaurantID = o.RestaurantID
GROUP BY r.RestaurantID, r.RestaurantName;

```

-- 20. Final Business Report

-- Which city should Swiggy focus on for expansion?

```

SELECT
    c.City,
    SUM(o.OrderAmount) AS Revenue
FROM Orders o
JOIN Customers c
ON o.CustomerID = c.CustomerID
GROUP BY c.City
ORDER BY Revenue DESC
LIMIT 1;

```

-- The city with the highest total revenue should be prioritized for expansion.

-- Which restaurant is the best-performing partner?

```

SELECT
    r.RestaurantName,
    SUM(o.OrderAmount) AS Revenue
FROM Orders o
JOIN Restaurants r
ON o.RestaurantID = r.RestaurantID
GROUP BY r.RestaurantName
ORDER BY Revenue DESC
LIMIT 1;

```

-- The restaurant with the highest total revenue.

-- Which cuisine should be promoted more?

```

SELECT
    r.Cuisine,
    COUNT(*) AS Orders
FROM Orders o
JOIN Restaurants r
ON o.RestaurantID = r.RestaurantID
GROUP BY r.Cuisine
ORDER BY Orders DESC
LIMIT 1;

```

-- The cuisine receiving the highest number of orders.

-- Who are the top 5 most valuable customers?

```

SELECT

```

```
c.CustomerName,  
SUM(o.OrderAmount) AS Total_Spending  
FROM Customers c  
JOIN Orders o  
ON c.CustomerID = o.CustomerID  
GROUP BY c.CustomerName  
ORDER BY Total_Spending DESC  
LIMIT 5;
```

```
-- Which delivery partner deserves a performance bonus?  
SELECT  
    DeliveryPartner,  
    COUNT(*) AS Successful_Deliveries  
FROM Orders  
WHERE OrderStatus='Delivered'  
GROUP BY DeliveryPartner  
ORDER BY Successful_Deliveries DESC  
LIMIT 1;
```

```
-- What is the average customer spending per order?  
SELECT  
    AVG(OrderAmount) AS Average_Order_Value  
FROM Orders;
```

```
-- Which restaurants need improvement based on ratings?  
SELECT  
    RestaurantName,  
    Rating  
FROM Restaurants  
WHERE Rating < 4.3;
```

```
-- What percentage of orders were cancelled?  
SELECT  
    ROUND(  
        COUNT(CASE WHEN OrderStatus='Cancelled' THEN 1 END)  
        *100.0  
        /  
        COUNT(*),  
        2  
    ) AS Cancelled_Percentage  
FROM Orders;
```

```
-- Which city has the highest average order value?  
SELECT  
    c.City,  
    AVG(o.OrderAmount) AS Average_Order_Value
```

```
FROM Orders o
JOIN Customers c
ON o.CustomerID = c.CustomerID
GROUP BY c.City
ORDER BY Average_Order_Value DESC
LIMIT 1;
```